Soham Sanjay Dandekar

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.66
Shriram Global School, Whitefield	Class XII (CBSE)	2024	95.20%
Ryan International School, Kundalahalli	Class X (ICSE)	2022	97.40%

PROJECTS

Huffman Text Compressor and Decompressor | C++

- Designed and implemented a complete **Huffman-based text compression system** using **min-heap-driven tree construction** for optimal prefix coding.
- Achieved efficient space utilization by packing **bit-level encoded data** into binary files, using **custom metadata** to handle non-byte-aligned streams and preserve exact bit length.
- Developed **modular encoder and decoder** components with separate mapping files to enable robust, **lossless decompression** of arbitrary text inputs.

Rare-Event Probability Estimator - (Python,NumPy)

- **Simulated rare-event probabilities** for extreme tails by generating 10^6 standard normal samples using the Box-Muller transform.
- **Implemented Importance Sampling** using a shifted proposal distribution to significantly increase sample density in the rare-event region.
- **Optimized statistical variance**, reducing estimator variance by multiple orders of magnitude compared to direct Monte Carlo simulation methods.
- **Built a custom random-number generation pipeline** by implementing the Box-Muller transform from scratch, converting uniform distributions into standard normal samples without relying on built-in library packages.

Producer-Consumer POSIX Implementation

- **Implemented the classic Bounded-Buffer problem** in C using fork() for multi-process creation and circular buffer indexing logic.
- **Managed process synchronization** via POSIX binary and counting semaphores (sem_t) to prevent race conditions during concurrent buffer access.
- **Architected shared memory infrastructure** using POSIX mmap(), allowing safe, asynchronous data transfer between isolated parent and child processes.

SKILLS

- **Languages:** C++, C, Python
- **Technical Skills:** Data Structures & Algorithms, NumPy, matplotlib, Git/GitHub, LaTeX
- **Tools:** VS Code, Jupyter Notebook, Git

RELEVANT COURSES

Data Structures and Applications, Algorithms, Operating Systems, Discrete Mathematics, Probability & Random Variables, Introduction to Programming, Numerical Methods

POSITIONS OF RESPONSIBILITY

Career Cell Coordinator | Office of Career Services

- **Planned career workshops, mock interviews, and guest lectures** to introduce students to emerging industry domains and trends.
- **Guided students on CV building and profile optimization** while creating domain-specific resources to bridge readiness gaps.

Web Head | Milan 2026

- **Leading the end-to-end web domain** to design and architect an engaging website for promoting festival events.
- **Monitoring and maintaining the website** to prevent downtime and ensure smooth performance during peak competition hours.
- **Implementing a real-time scoring system** and dynamic leaderboard for instantaneous, live event tracking.

Core Member, InFocus | Photography Club, IIT Hyderabad

- **Organized and led** campus photo walks and field tours, managing logistics and conducting hands-on technical sessions for members.

SCHOLASTIC ACHIEVEMENTS

- **Secured an SGPA of 9.65 in the 2nd semester**, ranking as one of the highest scores across the entire engineering branch.
- **Ranked in the top 5 of the academic cohort**, securing selective admission into the Artificial Intelligence specialization track.
- **Ranked 4th** in the Inter IIT Theme-Based Photography Competition at the Inter IIT Cultural Meet 2025, held at IIT Kanpur.